MILITARY SPECIFICATION

SOFTWARE QUALITY ASSURANCE PROGRAM REQUIREMENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Applicability. When referenced in the item specification, contract, or order, this specification shall apply to the acquisition of software (computer programs and related data & documentation) where the acquisition involves either software alone or software as a portion of a system or subsystem. This specification shall also apply to non-deliverable design, test, support, and operational software developed under the contract, unless specifically exempted. For purposes of this specification, the term software includes firmware.

1.2 Contractual Intent. This specification requires the establishment and implementation of a Software Quality Assurance (SQA) Program (hereafter referred to as the "Program") by the contractor. The purpose of the Program is to assure that software developed, acquired, or otherwise provided under the contract complies with the requirements of the contract. It is intended that the program be effectively tailored and economically planned and developed in consonance with, or as an extension of, the contractor's other quality assurance, administrative, and technical programs. The term "Program", as used herein identifies the collective requirements of the specification. The Program shall require periodic assessment and, where necessary, realignment of the Program to conform to changes in the acquisition program. The Program is subject to disapproval by the Government whenever it does not accomplish the requirements of this specification.

1.3 Relation to Other Contract Requirements. The contractor is responsible for compliance with all provisions of the contract and for furnishing specified software which complies with all the requirements of the contract. The SQA Program Plan shall reference other plans; e.g., configuration management, test, development, etc., specified under the contract and shall be compatible and consistent with them and not unnecessarily duplicate their provisions. If any inconsistency exists between the terms of the contract and this specifications, the Order of Precedence clause of the contract shall govern.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to Commander, US Army Computer Systems Command, ATTN: ACSC-TEC Stop C-100, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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2.0 APPLICABLE DOCUMENTS:

2.1 Amendments and Revisions. Whenever this specification is amended or revised subsequent to its contractually effective date, the contractor may follow, or authorize his subcontractors to follow, the amended or revised document provided no impact on schedule or increase in cost, price, or fee is required. The contractor shall not be required to follow the amended or revised document except as a formally authorized mofification to the contract. If the contractor elects to follow the amended or revised document, he shall notify the contracting officer in writing of this election. When the contractor elects to follow the provisions of an amendment or revision, he must follow them in full.

2.2 Ordering Government Documents. Copies of specifications, standards, and documentation required by contractors in connection with specific procurements may be obtained from the procuring agency, or as otherwise directed by the contracting officer.

3. REQUIREMENTS.

3.1 Software QA Program. Upon contract award, the contractor shall plan, develop, and implement a SQA Program which includes practices and procedures to assure compliance with all software requirements of the contract. The Program activities shall be a part of the management reporting system throughout the life of the contract. The contractor shall document the Program in the form of a SQA Plan (hereafter referred to as the "Plan") which meets the requirements of this specification. The Plan shall identify organizational responsibilities and authorities for its execution and the events critical to its implementation. The Plan shall also identify and make timely provisions for special needs (controls, tools, facilities, skills, etc) required for the Program and shall provide for detection, reporting, analysis, and correction of software problems and deficiencies. Contractor personnel performing quality functions shall have the responsibility, authority, and organizational freedom to evaluate software activities, identify problems, and initiate or recommend corrective action.

3.2 Software QA Program Requirements. The Plan shall address the following requirements:

3.2.1 Tools, Techniques, and Methodologies. The Plan shall identify the tools, techniques, methodologies and records to be employed in the performance of the work which will support QA objectives and describe how their use will augment or satisfy QA Program requirements. Examples include: Operations Research - Systems Analysis techniques, functional and performance requirements analysis, error analysis, software optimization tools, specification tracing, and coding conventions.

3.2.2 Computer Program Design. The Plan shall reference or document the procedures by which design documentation is reviewed to evaluate design logic, fulfillment of requirements, completeness, and compliance with specified standards. Design documentation shall be subjected to independent review prior to its release for coding.

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3.2.3 Work Certification. The Plan shall reference or document the contractor's procedures for formally approving or certifying the description, authorization, and completion of work performed under the contract. The Program shall require monitoring to assure compliance with these procedures.

3.2.4 Documentation. Documentation standards and programing conventions and practices to be used for all software shall be referenced or documented in the Plan. The Plan shall reference or document the procedures to be applied to assure compliance with standards, practices, and conventions and delivery of correct documentation and change information to the Government. In addition, the Plan shall provide for the independent review of documentation and designation of contractor approval authority.

3.2.5 Computer Program Library Controls. The Plan shall reference or document the contractor's procedures and controls for the handling of source code and object code and related data in their various forms and versions, from the time of their initial approval or acceptance until they have been incorporated into the final media. The objective of these controls is to ensure that different computer program versions are accurately identified and documented, that no unauthorized modifications are made, that all approved modifications are properly incorporated, and that software submitted for testing is the correct version.

3.2.6 Reviews and Audits. The Plan shall reference or document the contractor's procedures for preparation and execution of reviews and audits, for establishing the traceability of initial contract requirements through the successive baselines, and for ensuring that the reviews and audits are conducted in accordance with the prescribed procedures. The schedule for review and audits shall be referenced or stated in the Plan.

3.2.7 Configuration Management (CM). The Plan shall specify the relationships between the SQA and CM Programs and shall reference or document the procedures for assuring that the objectives of the CM program are being attained.

3.2.8 Testing. The Plan shall reference or document procedures for assuring the accomplishment of the following:

a. Analysis of software requirements to determine testability.

b. Review of test requirements and criteria for adequacy, feasibility, and traceability and satisfaction of requirements.

c. Review of test plans, procedures, and specifications for compliance with contractor and contractual requirements and to insure that all authorized and only authorized changes are implemented.

d. Verification that tests are conducted in accordance with approved test plans and procedures.

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e. Certification that test results are the actual findings of the tests.

f. Review and certification of test reports.

g. Ensuring that test related media and documentation are maintained to allow repeatability of tests.

h. The contractor shall ensure that support software and computer hardware to be used to develop and test software and hardware under the contract are acceptable to the Government.

3.2.9 Corrective Action. The Plan shall reference or document procedures which assure the prompt detection, documentation, and correction of software problems and deficiencies. Procedures shall include:

a. Documenting and reporting problems and deficiencies to appropriate management levels.

b. Analysis of data and examination of problem and deficiency reports to determine their extent and causes.

c. Analysis of trends in performance of work to prevent the development of noncompliant products.

d. Review of corrective measures to ensure that problems and deficiencies have been resolved and correctly reflected in the appropriate documents.

e. Analysis or review as otherwise provided for in the contract.

3.3 Subcontractor Control. The Plan shall reference or document the procedures to assure that all software acquired from subcontractors conforms to applicable requirements of the contract and this specification. When the Government elects to perform reviews at the subcontractor's facilities, such reviews shall not be used by contractors as evidence of effective control of quality of subcontractors by the contractor. It does not relieve the contractor of his responsibility for furnishing software that meets all contract requirements.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Contractor. Nothing specified herein relieves the contractor from the obligation to submit, to the Government for acceptance, end products that conform to all contract requirements.

4.2 Goverment Review at Contractor, Subcontractor, or Vendor Facilities. The Government reserves the right to review, at their sources, all products or services, including those not developed or performed at the contractor's facility, to determine the conformance of products or services with contract requirements.

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5. PREPARATION FOR DELIVERY. The Plan shall reference or document procedures for assuring integrity of software products during handling, storage, preservation, packaging, and shipping.

6. NOTES.

(The following information is provided solely for guidance in using this specification. It has no contractual significance).

6.1 Intended Use. This document will apply specifically to the acquisition of computer software where the acquisition involves either software alone, or software as a portion of a system or subsystem.

6.2 Ordering Data. The procuring activity should consider specifying the following:

6.2.1 Procurement Requirements.

a. Title, number, and date of this specification.

b. Software QA Program Plan. Consideration should be given to requiring the contractor to deliver a Software QA Program Plan in response to the invitation for bid, or request for proposal, or request for quotation and as a Contract Data Requirements List item (see 6.2.2). The Plan should define the methods and procedures which the contractor proposes to use in fulfilling the requirements of this specification. Note: The Software QA Program Plan may be included as part of other plans, see paragraph 1.3.

c. The application of this specification should be carefully tailored to meet the minimal essential needs of the acquisition.

d. Consideration should be given to citing current standards and specifications for configuration management, documentation, review, audit, development practices, work breakdown structures, etc.

e. Application of this specification to software maintenance contracts is encouraged.

f. Rapidly changing technology may require the acquiring activity to clarify use or application of the term "firmware".

6.2.2 Contract Data Requirements. All plans, documentation, and reports which are required to be delivered to the Government will be specified on a DD Form 1423, Contract Data Requirements List (CDRL) or authorized equivalent. The format, type of copy, number of copies, degree of detail required, delivery schedules, and purpose of submission should be specified on the CDRL.

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Custodians:

Army - AD

Air Force - 10

Navy - NM

Preparing Activity:

Army - AD

Project No. QCIC-0001

Review Activity:

Army - CR
ER
MI
SC
MD

Navy - NM
OM

Air Force - 01
05

Marine Corps - MC

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